BCSE204L Design and Analysis of Algorithms

Module-3: Naive String Matching

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SCOPE

Naive string matching algorithm

```
Begin
   patLen := pattern Size
   strLen := string size

   for i := 0 to (strLen - patLen), do
      for j := 0 to patLen, do
         if text[i+j] ≠ pattern[j], then
            break the loop
      done

      if j == patLen, then
         display the position i, as there pattern found
   done
End
```

Examples

- Text: $abcdefgh$, Pattern: def
- Text: $abcdabcabcdf$, Pattern: $abcdf$
- Text: $aaaaaaaf$, Pattern: $aaab$

Analysis

- Worst and average case: Time complexity of naive string matching is $O(m(n-m+1)$ where n is the string length and n is the pattern length
- Best case: When the first character in the pattern is not present in the string $\longrightarrow O(n)$

Thank you...